

End of Term Exam

Introduction to Probability and Statistics

Name: _____

ID: _____

Time allotted: 1 hour and 55 minutes

For examiner's use only

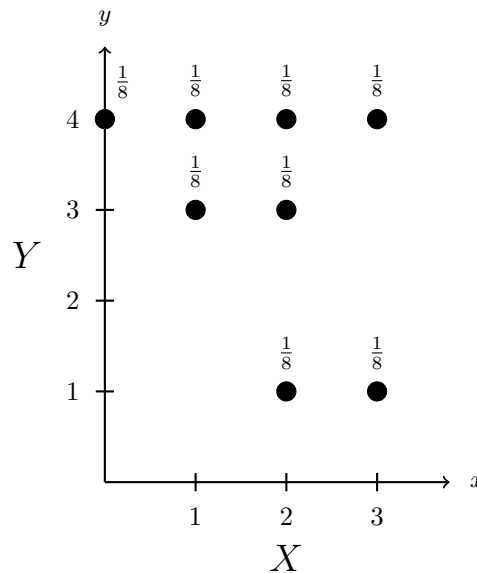
Question	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

Instructions:

- Arrive 10 minutes before the start of the exam.
- Mobile phones and other electronic devices must be **turned off** and put away **before** the start of the exam. You are allowed to use a scientific calculator (**Not** the one on your smart phone. Sharing of calculators is **not** allowed as well).
- **No** food or drink, other than water, is allowed in the examination room.
- You are **not** allowed to communicate with other candidates.
- You are **not** allowed to leave your seat unless granted permission by an invigilator.
- Assume where necessary and **justify** your assumptions.
- You are required to fill **all** answers in the provided **answer booklet**.
- You are **allowed** to bring a cheat sheet. It must be a **handwritten** A4 sheet of paper. You may write on both sides of the sheet. Photocopied cheat sheets are **not** allowed.
- The standard normal table has been provided on the last page.
- Good luck!

Question 1:

- (a) Let X be a Bernoulli random variable. X can take on the values 0 (failure) or 1 (success). It has probability of success, $p = 0.2$. Let Y be an independent Geometric random variable with parameter $p = 0.2$. Find the following:
- (i) $p_{X,Y}(2, 2)$. [2 points]
 - (ii) $p_{X|Y}(1|2)$. [4 points]
 - (iii) $p_{Y|X}(2|1)$. [4 points]
- (b) Consider the joint PMF of two random variables X and Y :



- (i) What is $E[Y|X = 2]$? [4 points]
- (ii) What is $E[X + Y]$? [6 points]

Question 2:

A metal sheet manufacturing plant produces sheets with a target thickness of 1 unit. The plant manufactures these metal sheets on two machines A and B . The thickness of the metal sheet produced by these machines can be modeled as a Gaussian Random Variable, Y , with mean equal to 1 unit. It is known that the new machine A has a Standard Deviation $\sigma_A = 0.004$ units while the old machine B has a Standard Deviation $\sigma_B = 0.005$ units. It is known that Machine A operates 90% of the time and Machine B operates 10% of the time (operated when machine A needs servicing).

The plant undertakes 100% inspection of the sheets before delivery and sheets with thickness, $0.99 \leq Y \leq 1.01$ units are acceptable and those outside these ranges are rejected.

- a) Calculate the percentage of sheets manufactured that are rejected. [15 points]
- b) The plant has recently hired an HU graduate who claims that she can reduce the percentage of sheets rejected by performing the servicing (of machine A) during off-hours and recommends an expensive upgrade of machine B to improve its standard Deviation $\sigma_B = 0.004$ units. What do you think her mathematical arguments could have been to justify these interventions? [5 points]

Question 3:

An HU student decides to play a game of lottery, where each ticket to play is PKR 500 for each game. The algorithm designed for winning/losing for each game is represented by Random Variable Z based on the following probabilities.

- Event A_1 - win a fixed amount, PKR 1,000 with probability 0.25.
- Event A_2 - win an amount that is uniformly distributed between PKR 600 to 1200 with probability 0.25.
- Event A_3 - you win nothing (PKR 0) with probability 0.5.

Therefore, the Random Variable Z is either 0 (with probability 0.5), or 1000 (with probability 0.25) or uniformly distributed between 600 to 1200 (with probability 0.25).

- a) Calculate and plot the Cumulative Distribution Function (CDF) for RV Z . Note - plot the CDF, clearly marking the exact CDF heights. **[12 points]**
- b) Find the probability, $P(Z \geq 500)$. **[4 points]**
- c) Now that you have completed the course on probability, what would your advice be to the student whether to play the game. Your advice should be based on mathematics (only) and whether the student would lose or win in the long run. **[4 points]**

Question 4:

A discrete random variable, X , takes three values, 0.2, 0.25, and 0.5 with equal probabilities. A Bernoulli random variable, Y , takes values +1 and -1, where the conditional probability, $P_{Y|X}(+1|x) = x$.

- a) Calculate and plot the PMF, $p_{Y|X}(+1|x) = x$. **[8 points]**
- b) Calculate and show the Joint PMF, $p_{X,Y}(x,y)$ for the DRVs X and Y . (Since the RV X takes on three values and the RV Y can take 2 values, you can use a two-dimensional grid to show the joint probabilities.) **[6 points]**
- c) Calculate the conditional probability, $p_{X|Y}(x|y = +1)$, for all the values of RV X . **[6 points]**

Question 5:

Radiologists often use CT scan intensity values to distinguish among different types of biological tissue. Suppose that for a particular region of the body, researchers have determined that:

- Soft tissue accounts for 50% of all pixels in that region of a CT scan, and the corresponding CT intensity values range from 25 to 45, with all values in this interval being equally likely.
- Dense tissue accounts for 35% of the pixels, and the corresponding CT intensity values range from 35 to 60, with all values in this interval being equally likely.
- Calcified tissue accounts for the remaining pixels, and its CT intensity values range from 40 to 90, with all values in this interval being equally likely.

This morning, you received the radiology report for a CT scan of a family member. For a particular pixel of clinical interest, the CT machine recorded an intensity of 40 units.

Based on this measured intensity, what is your diagnosis? (In other words, is the pixel most likely to be soft tissue, dense tissue, or calcified tissue)? **[20 points]**

(Note: Provide a justification of your answer using the rules, theorems, and results discussed in this course. Purely intuitive reasoning will not receive partial credit.)

	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4	0.99997	0.99997	0.99997	0.99997	0.99997	0.99997	0.99998	0.99998	0.99998	0.99998

Table 1: The standard normal table